

New MedViz Project Structure

1. Correcting and Completing the Rendering Pipeline (45% of the grade)

A partial and incorrect implementation of the volume rendering and lighting pipeline was unintentionally distributed as part of the project skeleton. Your task in this section is to **identify, correct, and complete** all mistakes and missing components in that implementation.

You must produce a fully functional, physically motivated emission–absorption volume renderer with lighting, entirely within the fragment shader.

Your corrections must address, at minimum:

A. Volume Rendering Corrections

- Proper ray–box intersection to compute entry and exit points.
- A correct emission–absorption integrator using:
 - front-to-back compositing,
 - premultiplied alpha,
 - early ray termination.
- A correctly chosen sampling step Δs , justified according to the Nyquist–Shannon theorem.
- Correct treatment of the transfer function.
- Stable and numerically correct opacity/color accumulation.

B. Lighting Corrections

- Compute voxel normals from the original scalar field.
- Correctly apply Lambertian shading at each sampling position.
- Shadow ray tracing must be fixed to:
 - originate at the current sample position,
 - compute enter/exit points for the secondary rays,
 - correctly accumulate opacity,
 - terminate early,
 - correctly sample disk-shaped area lights.
- Both visibility strategies described in the original specification must be implemented.

Your report must include in the METHOD section a detailed analysis of the errors, the corrections you implemented, and comparisons demonstrating the visual impact of these corrections.

2. New Section: Silhouette Enhancement for Volume Rendering (40% of the grade)

In addition to correcting the volume rendering pipeline, you must implement **an illustrative silhouette-enhancement technique** inside the fragment shader.

This technique highlights regions of the volume where the local surface normal is perpendicular to the viewing direction. These silhouettes improve shape perception, structure identification, and overall readability of the rendered volume.

What you must implement

At every ray-marching step:

- Compute the voxel normal (as used for Lambertian shading).
- Compute the viewing direction at the sample.
- Evaluate a silhouette weight w using a Fresnel-like term:

$$w = (1 - |N \cdot V|)^p$$

where $p > 1$ controls silhouette sharpness.

- Integrate this term into the rendering pipeline, affecting **color, opacity, or both**, in a visually meaningful way.
- Ensure the method remains stable across different datasets and transfer functions.

In the report, describe the proposed solution in the **METHOD section** and provide a clear comparison between traditional volume rendering and silhouette-enhanced rendering in the **RESULTS section**. In the **ANALYSIS section**, besides comment its impact in performance you should also discuss the perceptual benefits and the limitations of the approach.

Submission Instructions (unchanged)

You must still submit:

- The corrected fragment and vertex shaders.
- Your SimpleITK data-preparation tutorial (revised after peer feedback) [15% grade].
- A 4-page report following the Introduction → Related Work → Method → Results → Analysis → Conclusions structure.
- A updated GitHub Classroom repository with your full code. Remember evaluation will consider your pushes to the repo.
- A 5-minute demonstration video.

Deadline

January 19th at 08:00

Hard deadline, no late submissions will be accepted.